BME i9400 — Practice midterm (instructor copy)

For the review session · same format as the November 3 midterm · 100 points

Part A: 12 multiple-choice questions, 25 points. Part B: three of four problems, 25 points each. Every number below was checked by computation. The practice exam shares the midterm's format and topic list but none of its questions, scenarios, or numbers. It also exercises material that the midterm tests lightly — nested cross-validation, distances and cosines, and regression with an intercept — so students who work through both see the whole range.

Part A — Multiple choice (A1–A11: 2 points each; A12: 3 points)

	A1	A2	A3	A4	A5	A6	A7	A8	A9	A10	A11	A12
answer	b	d	c	b	c	a	a	c	d	b	a	d
meeting	2	2	4	4	6	7	8	9	11	12	9	12

- **A1:** $P(T{=}1) = 0.8 \times 0.1 + 0.1 \times 0.9 = 0.17$; PPV $= 0.08/0.17 \approx 0.47$. Distractor (c) is $P(T{=}1)$; (a) is the sensitivity.
- **A2:** $0.90/(1 - 0.95) = 18$.
- **A3:** $(1)(-2) + (2)(1) = 0$.
- **A6:** the weights are shrunk to zero and only the unpenalised intercept remains, which equals the mean of the training targets.
- **A10:** observed fraction $30/50 = 0.6 < 0.8$: the model's probabilities are too high for this group.
- **A11:** the AUC ignores prevalence; at 0.5% prevalence, even a small false positive rate produces many false positives per true positive, so the precision can be low.
- **A12:** the maximum of 20 noisy scores is biased upward; nested CV evaluates the whole selection procedure on data it never saw.

Part B — Problems (25 points each)

B1. Screening

(a) (6 pts) One point for each of the four interior cells; one each for the row totals and the column totals.

	test positive	test negative	total
cancer	45	5	50
no cancer	190	760	950
total	235	765	1,000

(b) (5 pts) PPV $= 45/235 \approx 0.191$ (3); NPV $= 760/765 \approx 0.993$ (2).

(c) (7 pts) $\mathrm{LR}^{+} = 0.9/0.2 = 4.5$ (2); $\mathrm{LR}^{-} = 0.1/0.8 = 0.125$ (2). Prior odds $= 5/95 = 1/19$; posterior odds after a negative test $= 0.125/19 \approx 0.0066$, so $P \approx 0.0065$ (2), which matches 5/765 from the table (1).

(d) (7 pts) The prior for the second test is the PPV of the first, 0.191, or odds 45/190 (3). Posterior odds $= 4.5 \times 45/190 \approx 1.07$, so $P \approx 0.52$ (4). By counting: of the 235 positives, $0.9 \times 45 = 40.5$ men with cancer and $0.2 \times 190 = 38$ without test positive again, giving $40.5/78.5 \approx 0.52$.

B2. Vectors and principal components

(a) (6 pts) $\|x\| = 5$ (2). $x - z = (-1, 1)$, so $\|x - z\| = \sqrt{2} \approx 1.41$ (2). $x \cdot z = 24$ and $\|z\| = 5$, so $\cos\theta = 24/25 = 0.96$ (2).

(b) (4 pts) Coordinate $x \cdot u = 2.4 + 2.4 = 4.8$ (2). Projection $4.8\,(0.8, 0.6) = (3.84, 2.88)$ (2).

(c) (9 pts) $\tilde{X}^\top \tilde{X} = \begin{bmatrix} 18 & 0 \\ 0 & 2 \end{bmatrix}$ and $C = \begin{bmatrix} 6 & 0 \\ 0 & 2/3 \end{bmatrix}$ (3). C is diagonal, so the principal components are the feature axes: PC1 $= (1, 0)$ with variance 6, PC2 $= (0, 1)$ with variance $2/3$ (4). Fraction explained by PC1: $6/(6 + 2/3) = 0.9$ (2).

(d) (6 pts) After standardisation each variance is 1 and the correlation is 0, so $C = I$ (2). Both eigenvalues equal 1, and the first component explains 0.5; every direction is equally a principal component (2). PCA depends on the scales of the features: on raw features, the one with the largest units dominates. Standardise first unless the units are comparable (2).

B3. Linear regression

(a) (4 pts) $X = \begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{bmatrix}$, $y = (1, 3, 5, 5)^\top$.

(b) (5 pts) $X^\top X = \begin{bmatrix} 4 & 6 \\ 6 & 14 \end{bmatrix}$ (3), $X^\top y = (14, 28)^\top$ (2).

(c) (7 pts) Determinant $= 4 \times 14 - 36 = 20$ (2). $w = \frac{1}{20} \begin{bmatrix} 14 & -6 \\ -6 & 4 \end{bmatrix} \begin{bmatrix} 14 \\ 28 \end{bmatrix} = \frac{1}{20} \begin{bmatrix} 28 \\ 28 \end{bmatrix} = (1.4, 1.4)$ (5).

(d) (6 pts) Predictions $1.4, 2.8, 4.2, 5.6$; residuals $-0.4, 0.2, 0.8, -0.6$ (3). $\sum r_i = 0$ and $\sum x_i r_i = 0 + 0.2 + 1.6 - 1.8 = 0$ (1). RMSE $= \sqrt{1.2/4} = \sqrt{0.3} \approx 0.55$ (2).

(e) (3 pts) (formula given) $\bar{y} = 3.5$, $\sum (y_i - \bar{y})^2 = 6.25 + 0.25 + 2.25 + 2.25 = 11$; $R^2 = 1 - 1.2/11 \approx 0.89$.

B4. Logistic regression and evaluation

(a) (9 pts) Odds $= e^z = e^{-1.1}(e^{0.69})^g \approx \frac{1}{3} \cdot 2^g$ (3). $g = 0$: odds $1/3$, $p = 0.25$; $g = 1$: odds $2/3$, $p = 0.40$; $g = 2$: odds $4/3$, $p = 4/7 \approx 0.57$ (2 each, using $p = \text{odds}/(1 + \text{odds})$).

(b) (9 pts) Predicted positive: the five women at 0.8. TP = 3, FP = 2, FN = 1, TN = 4 (1 each). Sensitivity 0.75; specificity $4/6 \approx 0.67$; precision $3/5 = 0.6$; accuracy 0.7 (5: 1 each, plus 1 for all four correct).

(c) (7 pts) At 0.2 the observed fraction is $1/5 = 0.2$: in agreement (3). At 0.8 it is $3/5 = 0.6$: the model overestimates the risk of its high group (3). The probabilities are not fully trustworthy as risks; a woman told "80%" has about a 60% risk. (Ten women is far too few to be sure — accept a remark to that effect.) (1)